

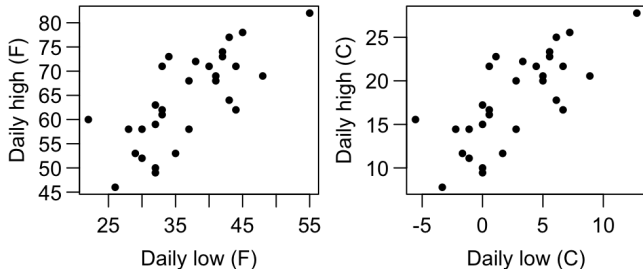
STAT 20000 Chapter 9

More About Correlation

Correlation r is NOT Affected by Change of Unit

Daily low and high temperatures for Boulder CO for the month of April 1996, in degrees Fahrenheit (left), and degrees Centigrade (right):

$$C = \frac{5}{9}(F - 32)$$

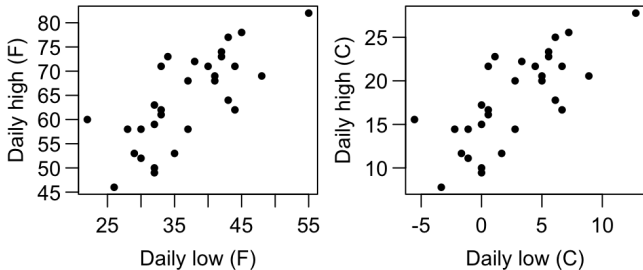


Are the correlation coefficients the same, or different?

Correlation r is NOT Affected by Change of Unit

Daily low and high temperatures for Boulder CO for the month of April 1996, in degrees Fahrenheit (left), and degrees Centigrade (right):

$$C = \frac{5}{9}(F - 32)$$



Are the correlation coefficients the same, or different?

The same, both $r = 0.74$. Why?

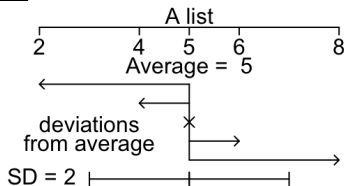
To see why, we are going to consider the effect of

- ▶ **translation** (adding the same number to each value of a variable),
- ▶ **scaling** (multiplying each value of a variable by the same positive number), and
- ▶ **reflection** (changing the sign of each value of a variable)

on the correlation coefficient.

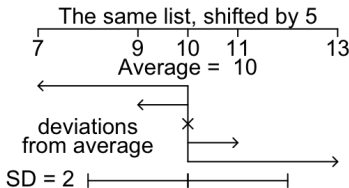
First we need to consider the effect of translation and scaling on standard units.

Recall Standard units are deviations from the average, expressed in unit



Standard Units

-3/2
-1/2
0
1/2
3/2



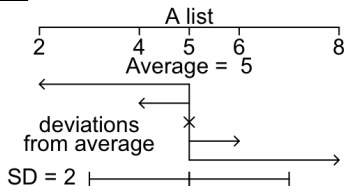
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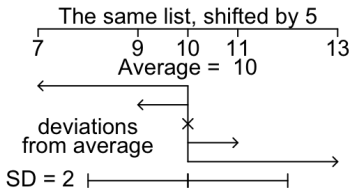
- ▶ the average ($\bar{x}$)
- ▶ the deviations from average ($x_i - \bar{x}$)
- ▶ the SD , and
- ▶ the *standard units* .

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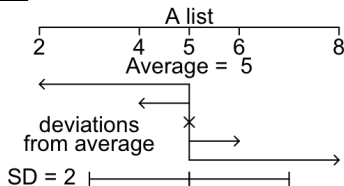
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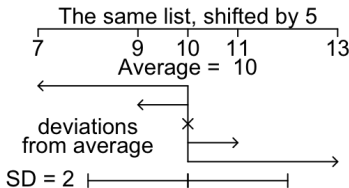
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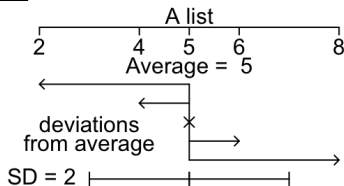
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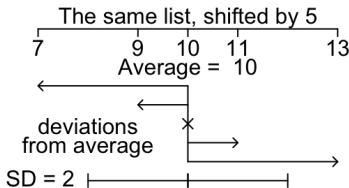
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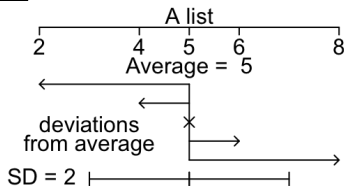
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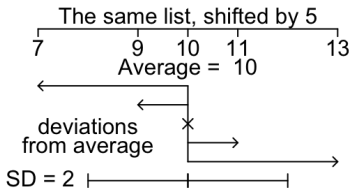
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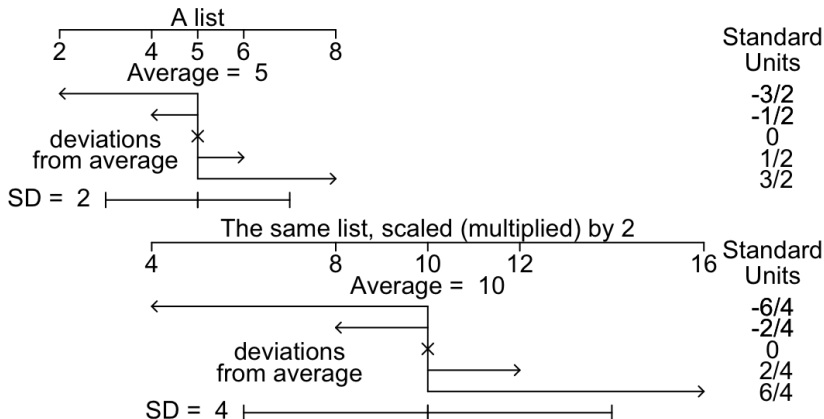


Standard Units

-3/2
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1/2
3/2

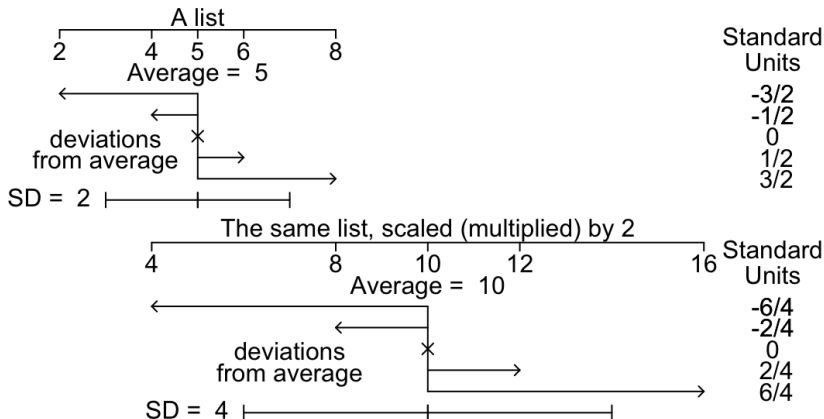
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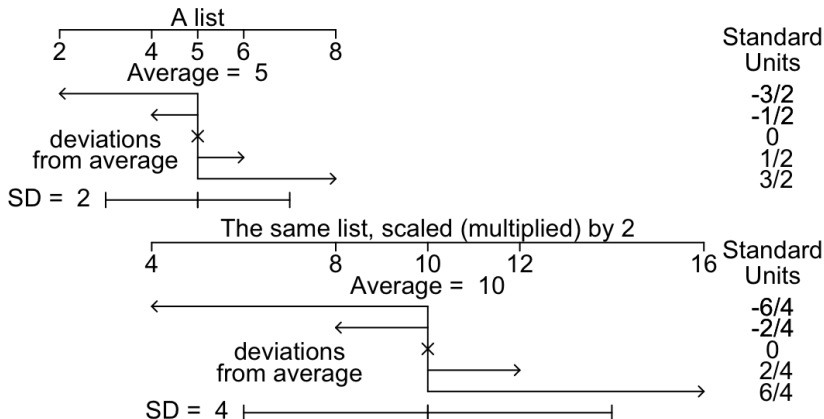
When all the entries on a list are multiplied by some number > 0 ,

- ▶ the average ($\bar{x}$)
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- ▶ the SD, but
- ▶ the *standard units*.



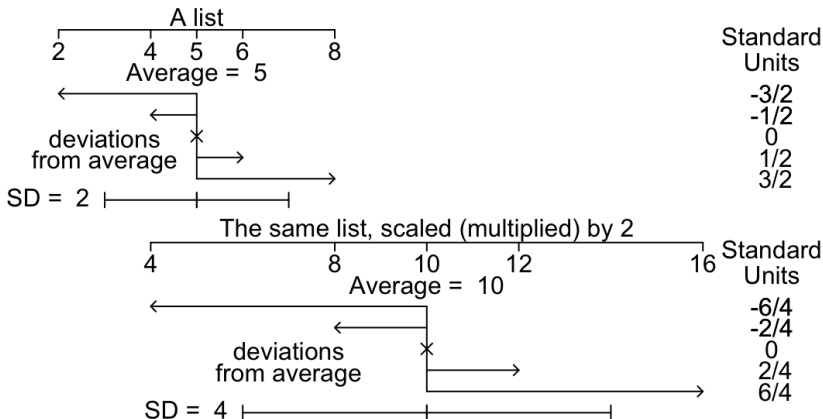
When all the entries on a list are multiplied by some number > 0 ,

- ▶ the average ($\bar{x}$) **gets scaled by that number**,
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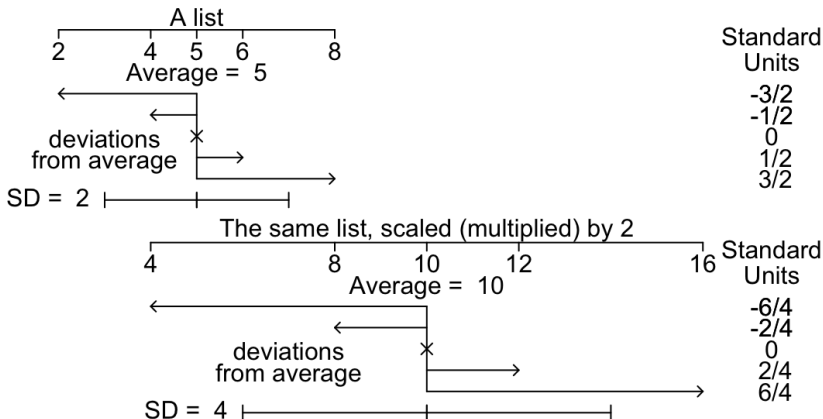
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When all the entries on a list are multiplied by some number > 0 ,

- ▶ the average ($\bar{x}$) **gets scaled by that number**,
- ▶ the deviations from average ($x_i - \bar{x}$) **get scaled**,
- ▶ the SD **gets scaled**, but
- ▶ the *standard units don't change*.

Translation and *Scaling* Have *no* Effect on the Correlation

since the correlation r between two variables x and y is computed from the formula

$$r = \text{average of } (x \text{ in standard units}) \times (y \text{ in standard units}).$$

No matter how often you translate and scale x and y , the standard units won't change, and neither will r .

Example:

The correlation between two sets of temperatures doesn't depend on whether the temperatures are measured in Fahrenheit or centigrade.

- Converting from Fahrenheit to centigrade involves a translation (subtract 32) followed by a scaling (multiply by $5/9$).

Example

Suppose 2 datasets are generated for cars of some brand:

- A. Number of miles a car can drive using one gallon of gas vs speed of head wind, in miles/hour;
- B. Number of kilometers a car can drive using one liter of gas vs speed of head wind, in km/hour;

Do x (distance) and y (wind speed) have the same correlation coefficient for these 2 datasets?

Example

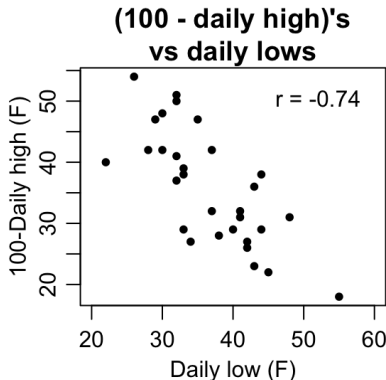
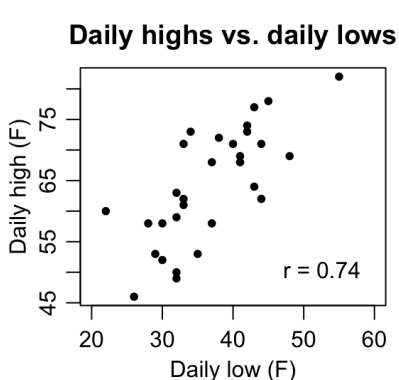
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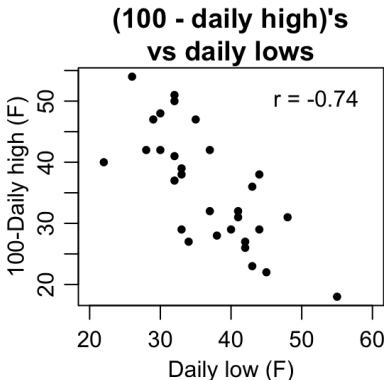
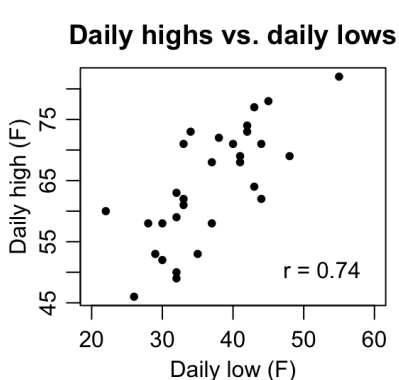
$$r_A = r_B$$

Watch out — if you multiply all the values of **ONE** of the variables by a **negative** number, you'll **change the sign** of the correlation



What if the values of **both** variables are multiplied by negative numbers?

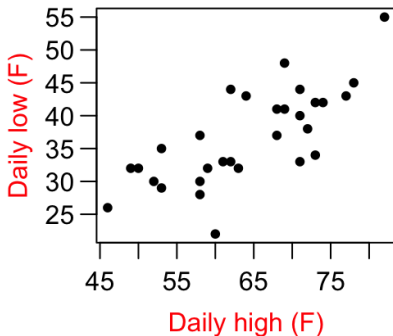
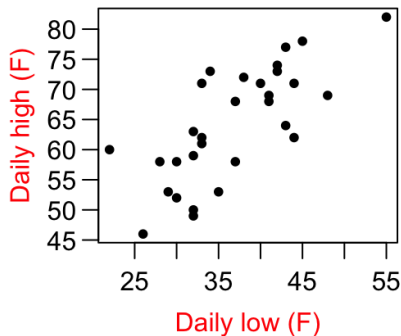
Watch out — if you multiply all the values of **ONE** of the variables by a **negative** number, you'll **change the sign** of the correlation



What if the values of **both** variables are multiplied by negative numbers? **r stay unchanged**

Correlation Does Not Distinguish X & Y

Q: Is the correlation between x and y the same as the correlation between y and x ?



A: The two correlations are **the same**. Why?

Reason: The correlation between x and y involves averaging the products

$(x \text{ in standard units}) \times (y \text{ in standard units})$, while

the correlation between y and x involves averaging the products

$(y \text{ in standard units}) \times (x \text{ in standard units})$.

But the order of the multiplications doesn't matter.

For example, if for some data point x is 3~SDs above average and y is 2~SDs below average, then

$$\begin{aligned} & (x \text{ in standard units}) \times (y \text{ in standard units}) \\ &= 3 \times -2 = -6 = -2 \times 3 \\ &= (y \text{ in standard units}) \times (x \text{ in standard units}). \end{aligned}$$

Switching the axes has no effect on the correlation coefficient.

	Shifting $x_i \rightarrow x_i + b$ for all i	Scaling $x_i \rightarrow ax_i$ for all i	Reflection $x_i \rightarrow -x_i$ for all i	Sensitive to extreme values?
Average $\bar{x}$	shifted by b $\bar{x} \rightarrow \bar{x} + b$	scaled by a $\bar{x} \rightarrow a\bar{x}$	change sign $\bar{x} \rightarrow -\bar{x}$	Yes
Median	shifted by b	scaled by a	change sign	No
SD	unchanged	scaled by $ a $ $SD \rightarrow a \times SD$	unchanged	Yes
IQR	unchanged	scaled by $ a $ $IQR \rightarrow a \times IQR$	unchanged	No
Standard Units	unchanged	multiplied by the sign of a	change sign	Yes

Let $r(x, y)$ be the correlation coefficient of two variables x and y .

$$r(x + a, y) = r(x, y)$$

$$r(ax, y) = \begin{cases} r(x, y) & \text{if } a > 0 \\ -r(x, y) & \text{if } a < 0 \end{cases}$$

$$r(y, x) = r(x, y)$$

Exercise: Correlation

Six data sets are shown below. The correlation is 0.8571 for (i), and 0.7857 for (ii). Can you find the correlations for (iii)(iv)(v)(vi)?

(i)		(ii)		(iii)		(iv)		(v)		(vi)	
x	y	x	y	x	y	x	y	x	y	x	y
1	2	1	2	2	1	2	2	1	4	0	1
2	3	2	3	3	2	3	3	2	6	1	2
3	1	3	1	1	3	4	1	3	2	2	3
4	4	4	4	4	4	5	4	4	8	3	4
5	6	5	6	6	5	6	6	5	12	4	5
6	5	6	7	7	6	7	5	6	10	5	6
7	7	7	5	5	7	8	7	7	14	6	7
$r =$	0.8571	0.7857									

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	x y	x y	x y	x y	x y	x y
	1 2	1 2	2 1	2 2	1 4	0 1
	2 3	2 3	3 2	3 3	2 6	1 2
	3 1	3 1	1 3	4 1	3 2	2 3
	4 4	4 4	4 4	5 4	4 8	3 4
	5 6	5 6	6 5	6 6	5 12	4 5
	6 5	6 7	7 6	7 5	6 10	5 6
	7 7	7 5	5 7	8 7	7 14	6 7
$r =$	0.8571	0.7857	0.7857			

(iii) is the same as (ii), with x and y switched, so r remains the same as (ii), 0.7857.

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x	y	x	y	x	y	x	y	x	y	x	y
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2	3	2	3	3	2	3	3	2	6	1	2
3	1	3	1	1	3	4	1	3	2	2	3
4	4	4	4	4	4	5	4	4	8	3	4
5	6	5	6	6	5	6	6	5	12	4	5
6	5	6	7	7	6	7	5	6	10	5	6
7	7	7	5	5	7	8	7	7	14	6	7
$r =$ 0.8571		0.7857		0.7857		0.8571					

(iv) comes from (i), by adding 1 to each x -value, so r is the same as (i), 0.8571.

Exercise: Correlation

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$r =$	<u>0.8571</u>		<u>0.7857</u>		<u>0.7857</u>		<u>0.8571</u>		<u>0.8571</u>		<u> </u>	

(v) comes from (i), by doubling each y -value, so r is the same as (i), 0.8571.

Exercise: Correlation

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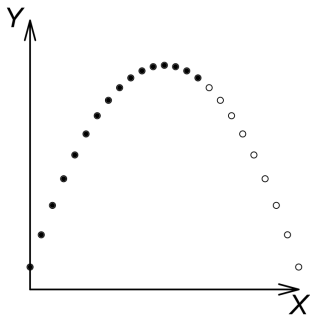
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	6	5	6	7	7	6	7	5	6	10	5	6
	7	7	7	5	5	7	8	7	7	14	6	7
$r =$	0.8571		0.7857		0.7857		0.8571		0.8571		1.0	

x and y in (vi) has a perfect linear relationship, $y = x + 1$. So $r = 1$.

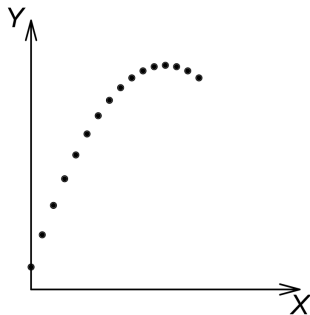
Correlation Doesn't Reflect Strength of Nonlinear Relations

Both scatter plots below show *perfect nonlinear* relations.

All points fall on the quadratic curve $y = 2 - x^2/2$.



$r = 0$ (why?)
(black + white dots)

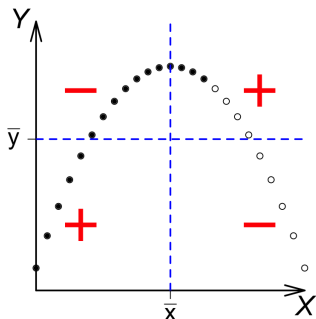


$r = 0.91$
(black dots only)

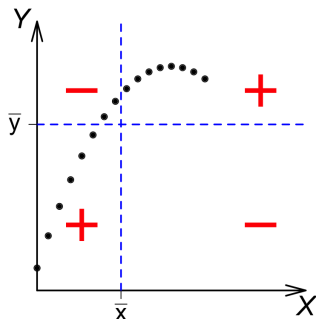
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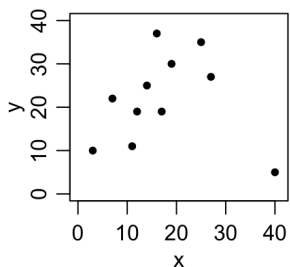
$r = 0.91$
(black dots only)

No matter how strong the relation, r is never 1 or -1 if it's nonlinear.

r can even be 0, like the plot above.

Correlation Is VERY Sensitive to Outliers

Sometimes a single outlier can change r drastically.

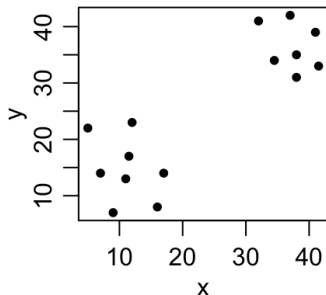


x	y
3	10
11	11
40	5
17	19
12	19
7	22
14	25
27	27
19	30
25	35
16	37

With the outlier (40, 5), the correlation is $r = 0.0044$.

Removing the outlier, the correlation becomes $r = 0.691$.

When Data Points Are Clustered ...



In the plot above, each of the two clusters exhibits a weak negative association ($r = -0.336$ and -0.323).

But the whole diagram shows a moderately strong positive association ($r = 0.849$).

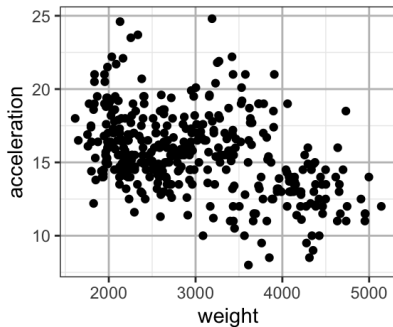
- ▶ This is an example of the **Simpson's paradox**.
- ▶ An overall r can be misleading when data points are clustered.
- ▶ Cluster-wise r 's should be reported as well.

Example: Auto Data

Auto is a data set containing information about 392 car models in the 1980s. The variables include

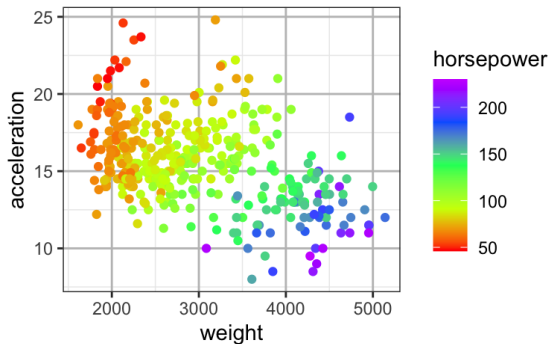
- ▶ acceleration: Time to accelerate from 0 to 60 mph (in seconds)
- ▶ horsepower: Engine horsepower
- ▶ weight: Vehicle weight (lbs.)

Example: Auto Data (2)



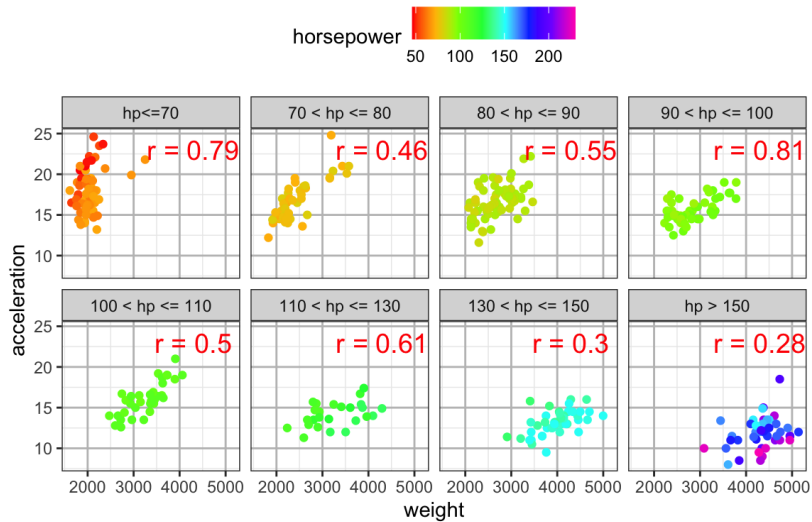
From the scatter plot, weight and acceleration are negatively correlated. Is that reasonable?

Example: Auto Data (3)



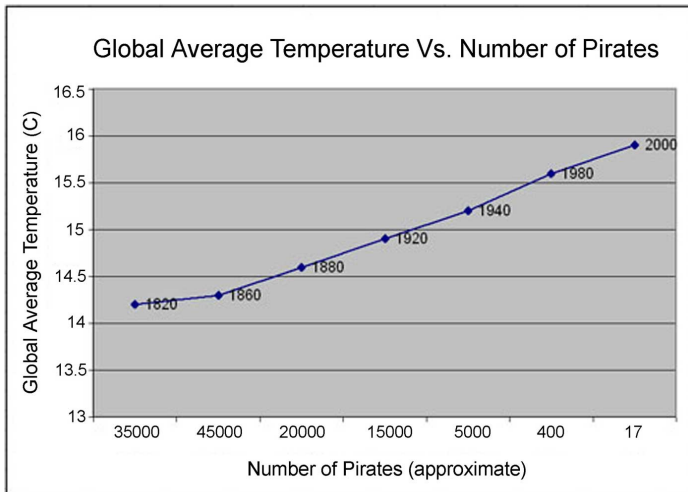
Consider car models of similar horsepower, are weight and acceleration positively or negatively correlated?

Example: Auto Data (4)



Correlation Indicates Association, *Not* Causation

STOP GLOBAL WARMING: BECOME A PIRATE



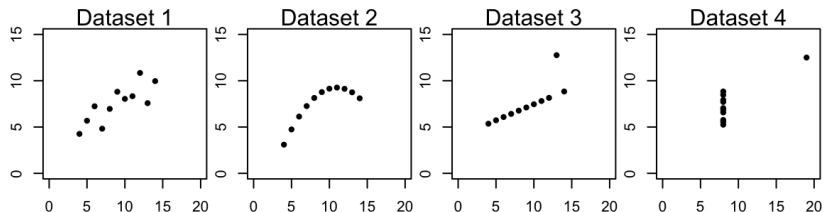
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Always Check the Scatter Plots (1)

The 4 data sets below have identical $\bar{x}$, $\bar{y}$, s_x , s_y , and r .

	Dataset 1		Dataset 2		Dataset 3		Dataset 4	
	x	y	x	y	x	y	x	y
	10	8.04	10	9.14	10	7.46	8	6.58
	8	6.96	8	8.14	8	6.77	8	5.76
	13	7.58	13	8.75	13	12.76	8	7.71
	9	8.81	9	8.77	9	7.11	8	8.84
	11	8.33	11	9.26	11	7.81	8	8.47
	14	9.96	14	8.10	14	8.84	8	7.04
	6	7.24	6	6.13	6	6.08	8	5.25
	4	4.26	4	3.10	4	5.36	19	12.50
	12	10.84	12	9.13	12	8.15	8	5.56
	7	4.82	7	7.26	7	6.42	8	7.91
	5	5.68	5	4.74	5	5.73	8	6.89
Ave	9	7.5	9	7.5	9	7.5	9	7.5
SD	3.16	1.94	3.16	1.94	3.16	1.94	3.16	1.94
r	0.82		0.82		0.82		0.82	

Always Check the Scatter Plots (2)

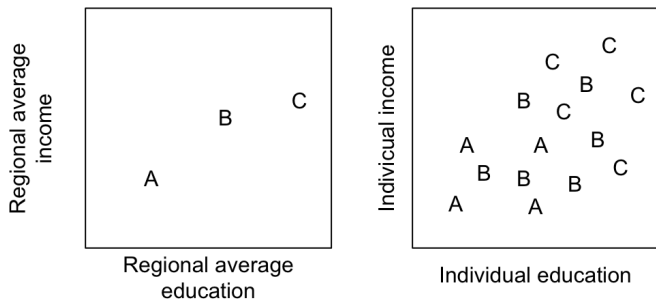


- ▶ For Dataset 2, y can be predicted exactly from x . But $r < 1$, because r only measures **linear** association.
- ▶ For Dataset 3, r would be 1 instead of 0.82 if the outlier were actually on the line.

The correlation coefficient can be misleading in the presence of outliers, multiple clusters, or nonlinear association.

Ecological Correlations

For each of 3 geographical regions in the U.S., a political scientist computes the average income and average educational level for the men aged 25–64 living in those regions. The correlation between regional average education and regional average income is 0.9. Does this give a fair estimate of the strength of the association between education and income for the men being studied?



- No. Taking averages eliminates spread, and gives a misleading impression of tight clustering.

Ecological Correlations (Cont'd)

In general, given a dataset of pairs of x and y values, divide the dataset into subgroups, and take the averages of x and y values within each subgroup. The correlation of x -average and y -average across the subgroups is **ecological correlation**.

(*Note*: this is different from correlation among variables from a population comprised of distinct clusters.)

Ecological correlations, which are based on rates or averages, tend to overstate the strength of associations.

Summary

- ▶ The correlation coefficient is a pure number, without units. It is not affected by:
 - ▶ interchanging the two variables;
 - ▶ adding the same number to all the values of one variable;
 - ▶ multiplying all the values of one variable by the same positive number.
- ▶ The correlation coefficient can be misleading in the presence of outliers or nonlinear association. Whenever possible, look at the scatter diagram to check for these problems.
- ▶ **Ecological correlations**, which are based on rates or averages, tend to overstate the strength of associations.
- ▶ Correlation measures association. But association does not necessarily show causation. It may only show that both variables are simultaneously influenced by some 3rd variable.